

Research Idea: Arp_220

ALMA Thesis Planner (automated pipeline)

Sign first, depth second: mapping the 650 μm continuum and the CO(4–3) line-to-continuum ratio across Arp 220’s two nuclei at ~ 30 pc

Data used (all of it, and only it): the single Band 8 member OUS ‘uid://A001/X1289/X1e6’ of ALMA project 2017.1.00735.S (PI Rangwala; public since 2019-11-22), one pointing on Arp 220 at RA, Dec = 233.73843, +23.50322; spectral windows 451.9–453.8 / 453.6–455.6 / 463.8–465.8 / 465.6–467.6 GHz; tabulated angular resolution 0.086'' (≈ 30 pc at Arp 220’s distance, per the archive annotation), native velocity resolution 0.746 km/s, t_int 302 s, line sensitivity 8.336 mJy/beam @10 km/s, continuum sensitivity 0.2723 mJy/beam. QA2 image products and pipeline-calibrated measurement sets from the ALMA Science Archive. The CO J=6–5, CO J=3–2, ^{13}CO 4–3 and ^{13}CO 6–5 data named in the proposal abstract belong to other, unlisted member OUS and are **not** in this dataset; multi-line excitation modeling and the $^{12}\text{C}/^{13}\text{C}$ isotopic ratio are explicitly out of scope.

The line and its stated provenance: the target transition is ^{12}CO J=4–3, rest frequency 461.041 GHz, which the archive annotation places at ≈ 452.8 GHz *assuming* Arp 220’s systemic redshift $z = 0.018126$ ($cz = 5434$ km/s), putting it inside the 451.9–453.8 and 453.6–455.6 GHz windows. That line list was generated from general knowledge and must be verified against Splatalogue, with the sky-frequency placement re-checked against the actual MS header, before any imaging. The 463.8–465.8 and 465.6–467.6 GHz windows are annotated only as *tentative* candidates for high-J HC_3N (J=51–50, rest ≈ 464.0 GHz; J=52–51, rest ≈ 473.1 GHz), and the annotation is itself agnostic as to whether these would be redshifted or near-rest-frequency serendipitous detections — that identification is not established. Those windows are used here as additional continuum only; any HC_3N feature is reported as a tentative, clearly-flagged secondary result.

Headline deliverable: a spatially and spectrally resolved characterization of (a) the 650 μm continuum of the east and west nuclei — positions, deconvolved sizes, peak and integrated flux densities — and (b) the **line-to-continuum ratio** $R(v) = I_{\text{line}}(v)/I_{\text{cont,local}}$ of CO(4–3) across those nuclei, at the finest spatial scale the data actually support. Whether $R < 1$ (absorption) or $R > 1$ (emission), and where the sign changes, is a *measurement this thesis makes*, not an assumption it starts from.

Sign is not assumed. The proposal abstract’s “deep CO absorption” statement concerns CO 6–5 and 3–2 in other member OUS. That CO(4–3) also goes below the continuum here is an inference, and the plan branches on it explicitly rather than discovering the problem in week 9:

- **Absorption branch** ($R < 1$ over a nucleus or a set of positions): report apparent absorption depth $D(v) = -\ln R(v)$, its peak, the absorption-weighted velocity centroid relative to $cz = 5434$ km/s, the absorption FWHM, and PV cuts along each nucleus’s major axis.
- **Emission branch** ($R > 1$): report integrated line flux, moment-0/1/2 maps, and PV cuts